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St. Joseph's College of Arts & Science (Autonomous)
St. Joseph's College Road, Cuddalore – 607001
B.Sc Degree Examination - APRIL 2021

MT204S – NUMERICAL METHODS

Time : 3 hrs **Max Marks :75**

SECTION – A (5X2=10)

Answer **ALL** Questions

1. Write Gregory –Newton forward interpolation formula.
2. Write Bessel's formula.
3. Write inverse interpolation formula.
4. Solve the equation $x + y = 2$ and $2x + 3y = 5$ by Gauss elimination method.
5. Write the formula for Simpson's three-eighth rule.

SECTION – B (3X5=15)

Answer any **THREE** Questions

6. Represent the function $f(x) = x^4 - 12x^3 + 24x^2 - 30x + 9$ and its successive differences in factorial notation.
7. Using Bessel's formula, find $f(25)$ given $f(20) = 2854, f(24) = 3162, f(28) = 3544, f(32) = 3992$
8. Find $f(8)$ from the following table

x	4	5	7	10	11	13
y	48	100	294	900	1210	2028

9. Using Gauss-Jordan method, solve
$2x_1 + 4x_2 + x_3 = 3,\ 3x_1 + 2x_2 - 2x_3 = -2,\ x_1 - x_2 + x_3 = 6.$

10. Evaluate $\int_0^1 \frac{dx}{1+x^2}$, Using Trapezoidal rule with $h = 0.2$ Hence determine the value of π.

SECTION - C (5X10=50)

Answer **ALL** Questions

11. a) Apply Newton's backward formula to find a polynomial of degree 3 which includes the following x, y pairs

x	3	4	5	6
y	6	24	60	120

(or)

b) From the following data, estimate the number of person earnings weekly wages between 60 and 70 rupees

Wages (in Rs.)	Below 40	40 – 60	60 – 80	80-100	100 – 120
No. of person (in thousand)	250	120	100	70	50

12. a) Use Gauss's backward formula, find $\sin 45^o$ from the following table

x^o	20	30	40	50	60	70
$\sin x^o$	0.34202	0.502	0.64279	0.76604	0.86603	0.93969

(or)

b) Using Bessel's formula, find the value of y at $x = 1.95$ given that

x	1.7	1.8	1.9	2.0	2.1	2.2	2.3
y	2.979	3.144	3.283	3.391	3.463	3.997	4.491

13. a) Using Lagrange's formula, find the form of the function $f(x)$ given that

x	0	2	3	6
$f(x)$	659	705	729	804

(or)

b) Find the value of x corresponding $y = 100$ from the table.

x	3	5	7	9	11
y	6	24	58	108	174

14. a) Solve the systems of equations $8x - y + z - 18 = 0, 2x + 5y - 2z - 3 = 0, x + y - 3z + 6 = 0$, using Gauss – Seidal iterative method.

(or)

b) Find the inverse of the matrix $\begin{pmatrix} 3 & 2 & 4 \\ 2 & 1 & 1 \\ 1 & 3 & 5 \end{pmatrix}$ using Gaussian method.

15. a) Using Improved Euler's method find y at $x = 0.1$ and $x = 0.2$ given $\frac{dy}{dx} = y - \frac{2x}{y}$, $y(0) = 1$.

(or)

b) Using Runge-Kutta method of fourth order solve $\frac{dy}{dx} = \frac{y^2 - x^2}{y^2 + x^2}$ with $y(0) = 1$ at $x = 0.2$.

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